Engineered MoO_3 Thickness Control for Area-Selective Chemical Vapor Deposition of Two-Dimensional MoS_2

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Introduction

2D semiconducting transition metal dichalcogenides (TMDs) show promise for device miniaturization, but challenges in scalability, uniformity, and substrate compatibility hinder commercial viability [1-7]. We address these issues with a novel area-selective synthesis method for high-quality TMD layers. Using MoS_2 as an example, we demonstrate transfer-free synthesis of MoS_2 arrays using patterned MoO_3 seeding. We introduce precursor thickness engineering for MoS_2 chemical vapor deposition, in which molybdenum oxide precursors are thinned by wet etching to an optimal thickness. Quantitative atomic force microscopy reveals power-law dissolution kinetics, providing a time-controlled knob that links precursor thinning to MoS_2 domain size and morphology. Using this engineered seed layer, we further optimize sulfur saturation, carrier-gas flow, and molybdenum source temperature to achieve self-limited, diffusion-controlled lateral growth and $MoS_2/MoO_x/Si$ heterostructures suitable for future device integration.

Material Synthesis

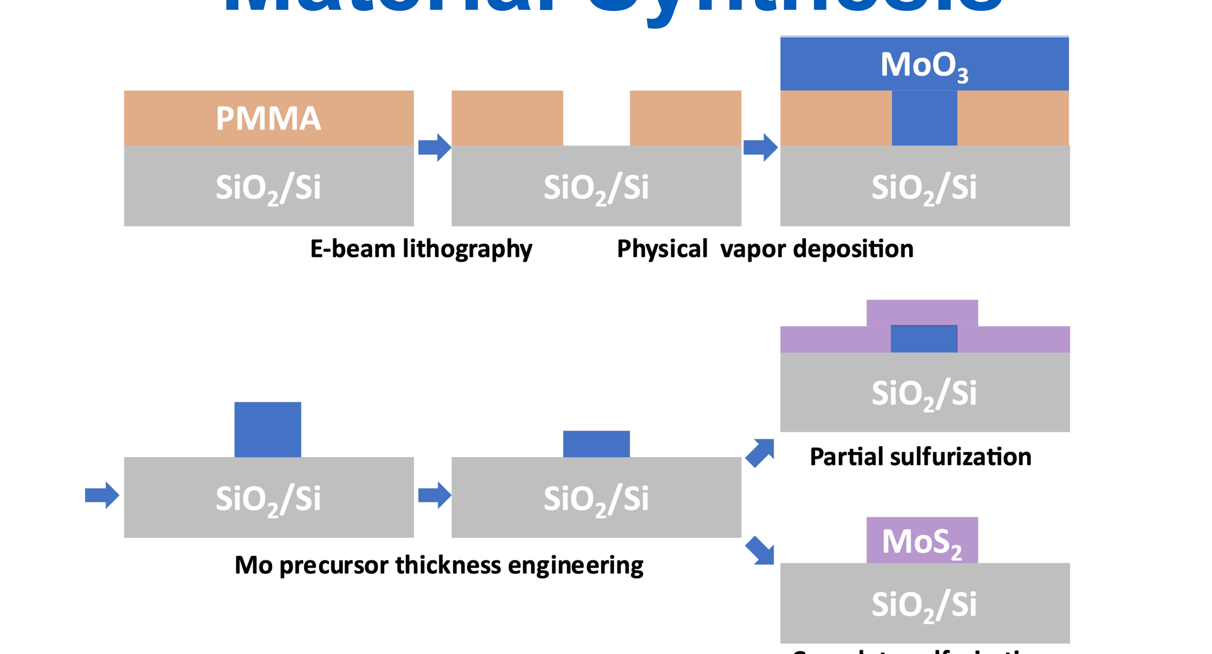
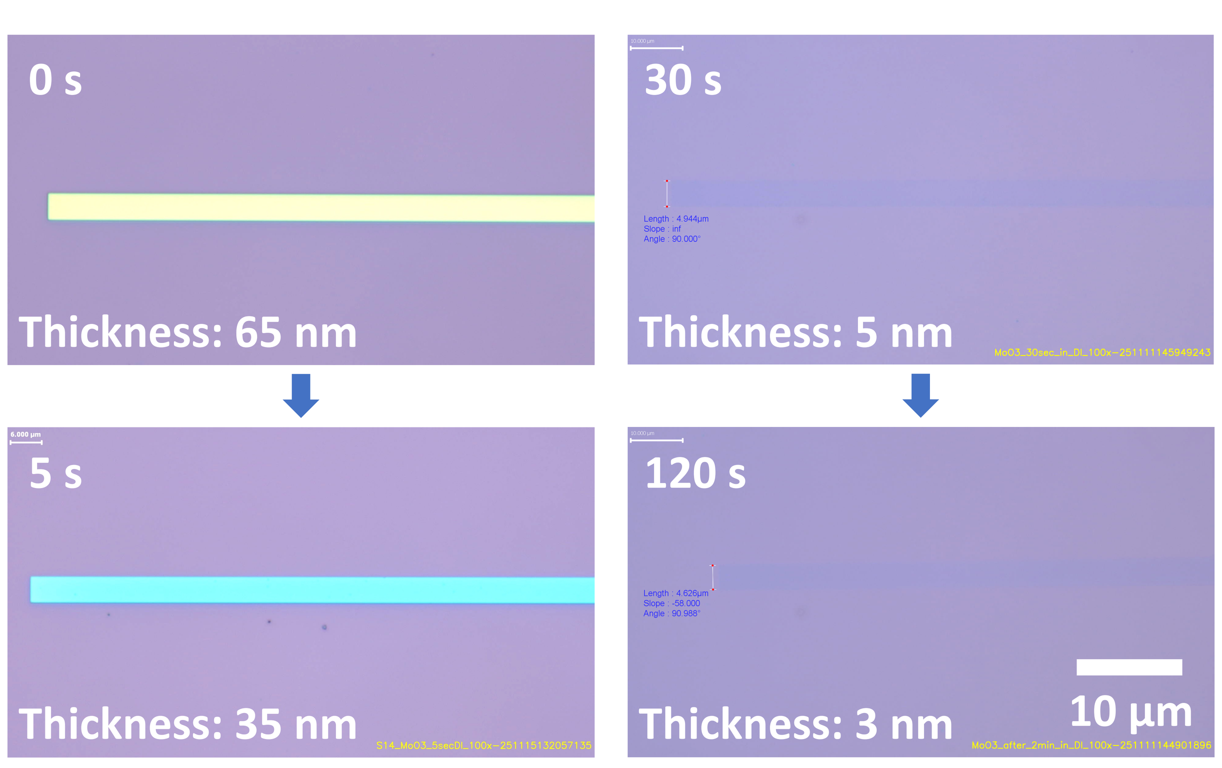


Fig. 1 Schematic flow for area-selective MoS_2 synthesis
The sequence includes polymethyl methacrylate coating and E-beam lithography patterning, followed by physical vapor deposition, acetone lift-off and isopropyl alcohol rinse, Mo precursor thickness engineering, and final sulfurization via chemical vapor deposition (CVD).



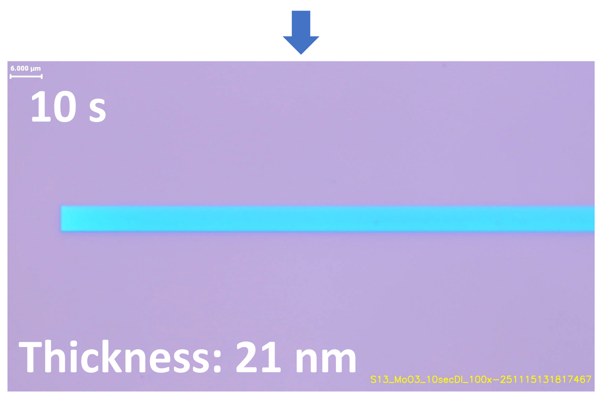


Fig. 2 Optical micrographs of precursor thinning during wet etching.
The thickness decreases from 65 nm to 3 nm over 120 sec, with clear color contrast.

Material Characterization

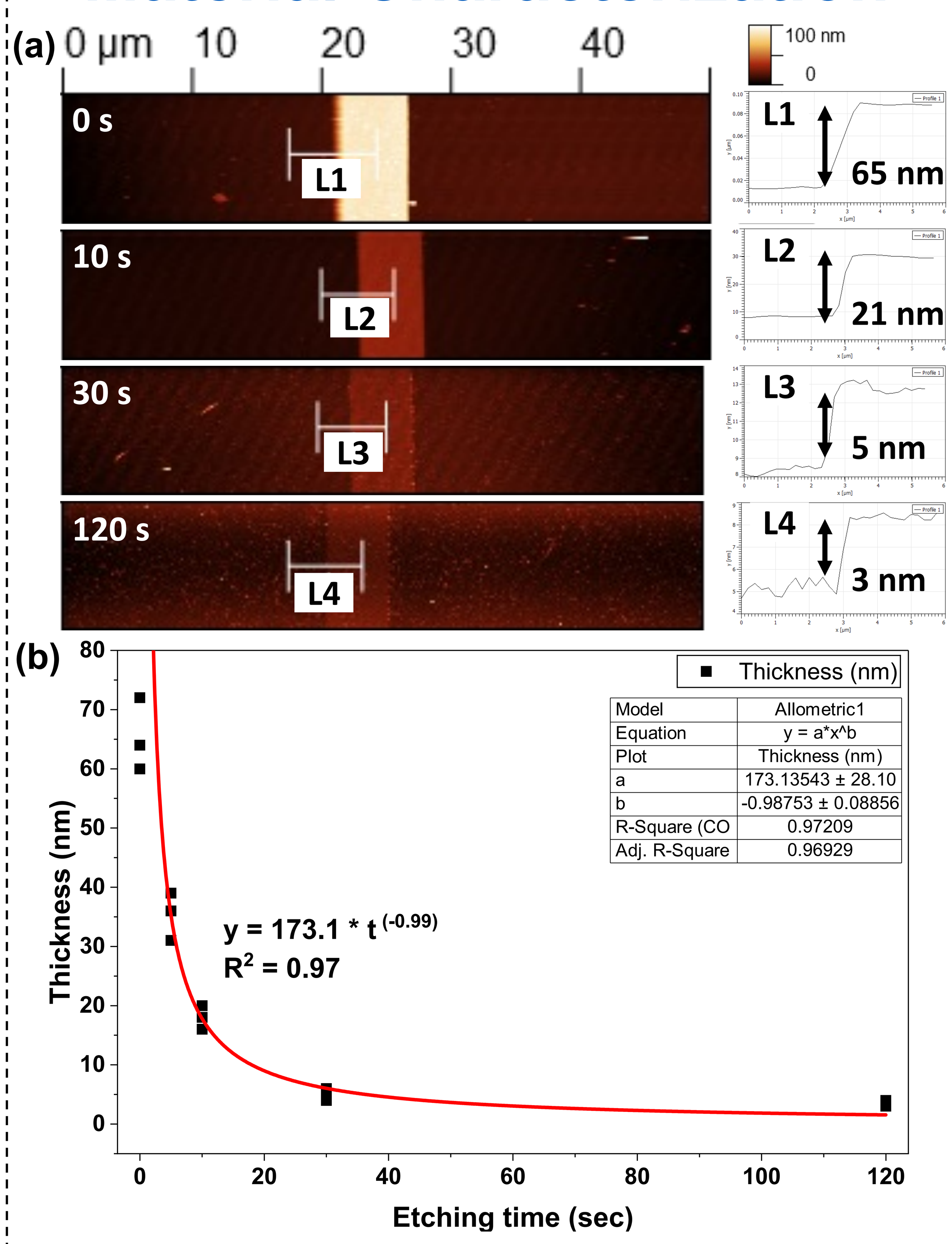


Fig. 3 Atomic Force Microscopy (AFM) height maps and line profiles confirming precise MoO_3 thickness reduction during wet etching.
(a) Height maps and line profiles at 0, 10, 30, and 120 sec etching times.
(b) Thickness vs. etching time with power-law fit $h = 173.1 \times t^{-0.99}, R^2 = 0.97$, indicating dissolution-limited kinetics thicknesses control.

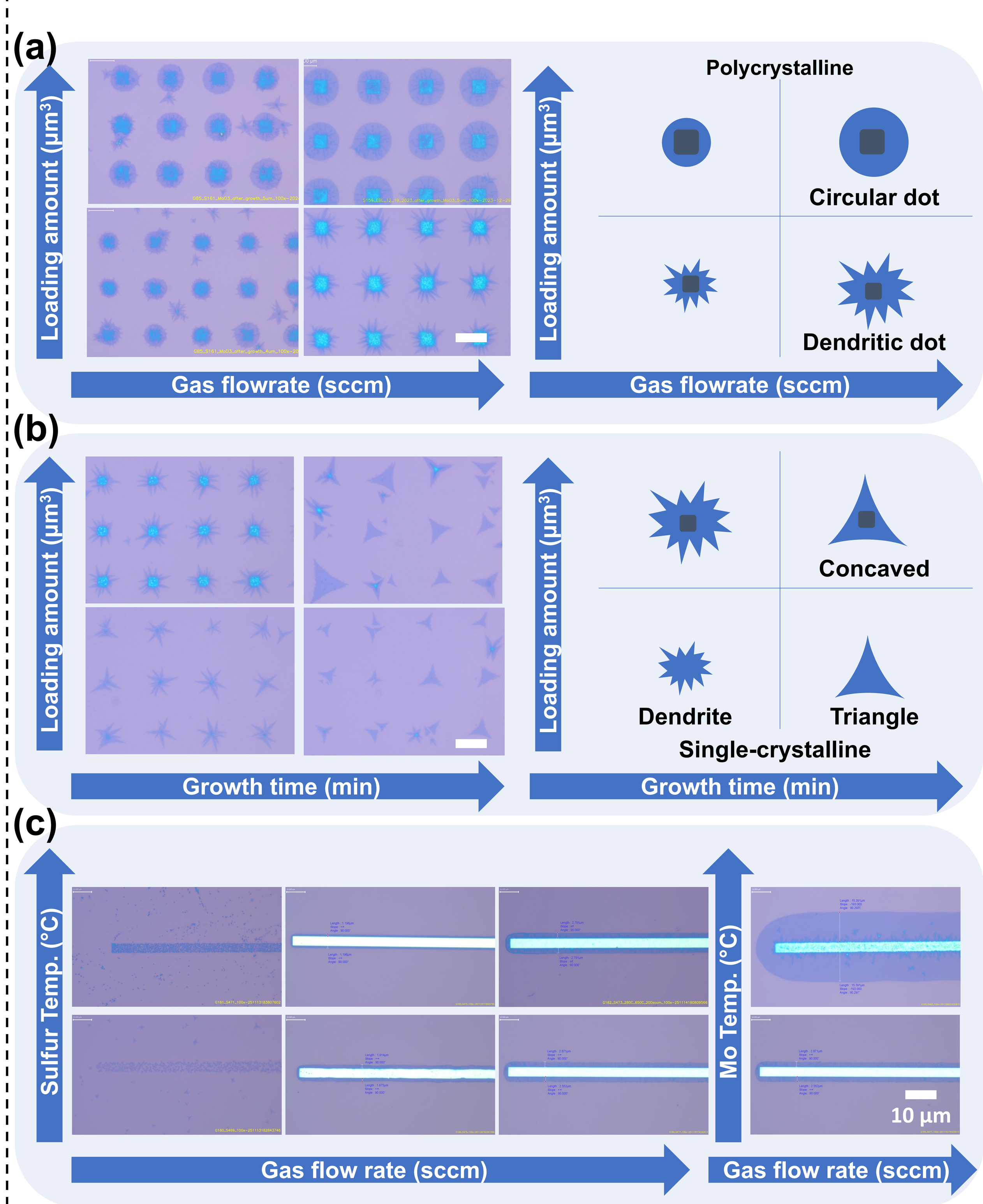


Fig. 4. CVD parameter space exploration and MoS_2 growth evolution.
(a) Mo precursor amount vs. gas flow rate.
(b) Mo loading amount and growth time: optimized for single-crystalline MoS_2.
(c) Temperature and gas flow rate tune the MoS_2 domain width from about 2 μm up to 10 μm.

Material Characterization

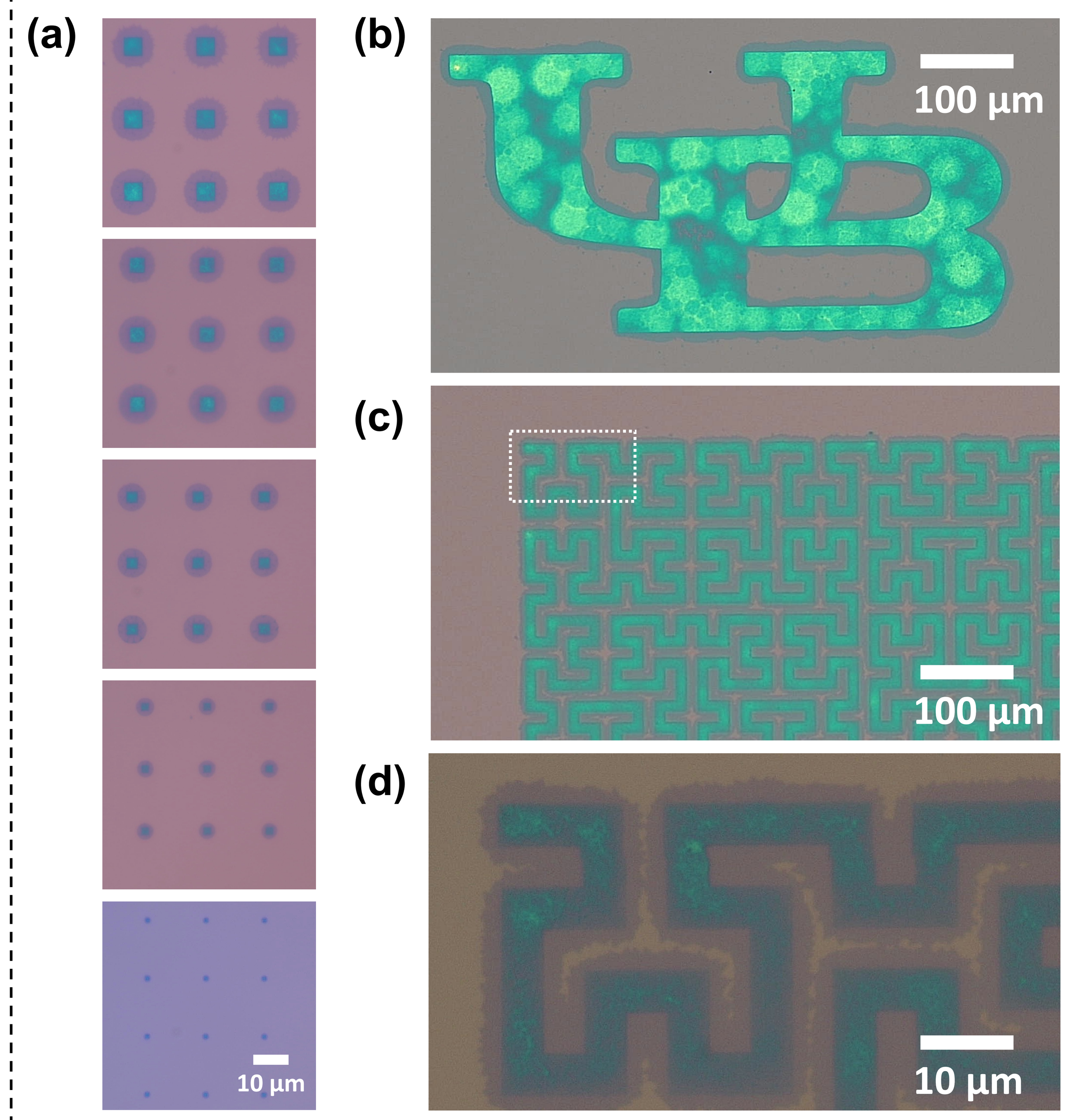


Fig. 5 Optical micrographs of patterned MoS_2.
(a) Lateral size tuning of polycrystalline MoS_2 islands from 5 μm to 1 μm via precursor loading and process control. (b–d) UB text and Hilbert-curve patterns demonstrating area-selective synthesis with high pattern fidelity.

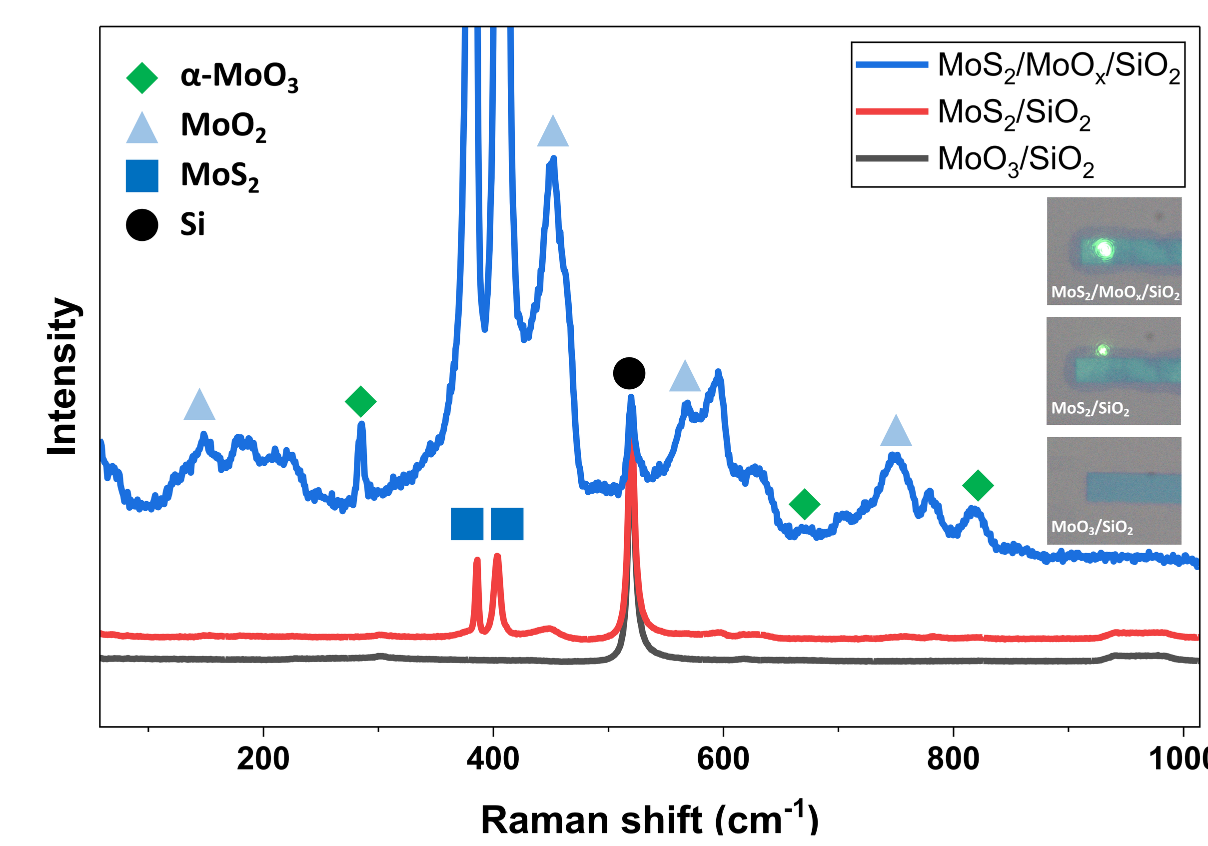


Fig. 6 Raman spectra of $MoS_2/MoO_x/SiO_2$.
Characteristic MoS_2 modes (E_{2g}~385 cm^{-1}, A_{1g} ~407 cm^{-1}), together with α-MoO_3 peaks at 289, 666, 819 cm^{-1} and MoO_2 peaks at 148, 457, 569, 741 cm^{-1}, confirm MoS_2 formation and crystallization of the interfacial MoO_x layer.

Conclusion

In this work, we developed a thickness-engineered, area-selective CVD route for MoS_2 that advances location-on-demand synthesis of 2D semiconductors. This approach enables time- and cost-efficient growth of channel arrays at predefined sites, while controlled sulfurization converts the thickness-engineered precursors into functional $MoS_2/MoO_x/Si$ heterostructures. Our methods combined precursor-thickness engineering and CVD-process optimization strategy can be extended to other transition metal dichalcogenides and heterostructures, offering a general pathway toward integrated 2D electronics and nanoelectronic systems.

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References

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